

Machine Learning & Data Visualization Lab

Assignment

Dataset Provided: messy_ecommerce_data.csv

Objective: The provided e-commerce dataset has been intentionally corrupted with various real-world data quality issues. Your objective is to use Python and the pandas library to systematically clean the dataset, transforming it into a structured, error-free format suitable for a multi-class classification model.

Task 1: Handling Missing Values (Cleaning Empty Cells)

Approximately 5% of the values in most columns have been randomly replaced with NaN.

Perform the following operations:

- Load the dataset and calculate the exact total and percentage of missing values for every column in the dataset.
- For categorical columns missing data, apply Mode imputation or fill with a placeholder string (e.g., "Unknown").
- For numerical columns missing data, decide between Mean or Median imputation and apply your chosen method. Add a brief comment in your code explaining why you chose one over the other.
- Verify that no missing values remain in the dataset after your operations.

Task 2: Standardizing Data Formats (Cleaning Wrong Format)

Several columns contain data that is technically present but structurally incorrect for machine learning algorithms. Standardize these formats:

- **Transaction_Date:** The column features a mix of standard YYYY-MM-DD and non-standard DD/MM/YYYY formats, alongside invalid string inputs like "Not a date". Use pandas datetime functions to convert all valid entries into a uniform YYYY-MM-DD datetime object. Convert the invalid strings into NaT (Not a Time) and then drop those specific rows.
- **Transaction_Amount:** This column is currently being read as an object/string because it incorporates float values mixed with random \$ signs and "Error" text strings. Strip the \$ signs, replace the "Error" strings with NaN, and convert the entire column to a float64 datatype.
- **Is_Active:** This column has mixed booleans with string formats like "Yes" and "No". Map these strings to their proper boolean equivalents (True for "Yes", False for "No") so the column contains strictly boolean values.

Task 3: Correcting Invalid Entries (Cleaning Wrong Data)

Some columns contain data of the correct type, but the values violate logical boundaries or formatting rules. Fix the following anomalies:

- **Customer_Age:** The dataset contains impossible biological ages, including negative numbers (-10) and extreme outliers (150, 999). Filter the dataset to either drop rows where the age is strictly outside the logical range of 18 to 100, or replace those specific outliers with the median age of the dataset.
- **Customer_Rating:** The rating is strictly a 1-5 scale, but includes out-of-bounds metrics like 0, 6, 10, and 99. Use a clipping function or conditional replacement to ensure no rating falls outside the 1.0 to 5.0 range.
- **City:** The categorical text introduces inconsistent casing (e.g., lowercase, UPPERCASE) and extra leading/trailing whitespaces. Apply string manipulation functions to strip all invisible whitespaces and convert all city names to standard Title Case (e.g., "new york " becomes "New York").

Task 4: Removing Duplicates

To prevent data leakage during model training and testing, redundant data must be removed.

- Identify how many exact duplicate rows exist in the dataset (Hint: there are exactly 500 scattered throughout).
- Drop these exact duplicate rows, keeping only the first occurrence of each observation.
- Print the final shape (number of rows and columns) of your cleaned dataframe.

Submission Guidelines

- Save your completed code in a Jupyter Notebook (.ipynb) or Google Colab file.
- Ensure all code cells are executed so the outputs are visible.
- Export the final cleaned dataframe to a new CSV file named `cleaned_ecommerce_data.csv` and submit it alongside your code.